

Alvin Ng Wei Sing

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EDUCATION

Singapore University of Technology and Design

Sep 2024 - May 2028 (Expected)

Bachelor of Engineering (Computer Science and Design)

GPA: 5.00/5.00

SUTD Global Distinguished Scholarship

Honours List AY24/25

Notable Coursework: Computation Structures (A+), Data Driven World (A+), Algorithms (A), Information Systems & Programming (A)

EXPERIENCE

Software Engineering Intern | Pettichat Tech Limited, Hangzhou, China

Oct 2025 - Dec 2025

Python, PyTorch, Librosa

- Developed a machine learning pipeline to classify 10 pet vocalisation categories with 80% accuracy by leveraging audio feature extraction with Librosa and deep learning models in PyTorch
- Reduced model latency by 50% by implementing knowledge distillation techniques to compress neural network architecture for edge deployment
- Built a data preprocessing pipeline for 3000 audio samples by converting to mel spectrogram representations

Undergraduate Researcher | Singapore University of Technology and Design, Singapore

Feb 2025 - Aug 2025

Python, PyTorch

- Implemented and evaluated versions of federated learning algorithms using PyTorch to study system performance under different training configurations
- Investigated clustering-based optimisation techniques for efficient machine learning systems

Engineering/Project Intern | MLS Logistic Services, Singapore

May 2024 - Aug 2024

Visual Basic

- Reduced invoice processing time by 90% by redesigning and automating company workflow in Visual Basic
- Automated inventory accounting processes to eliminate repetitive manual tasks for 20 warehouse staff
- Designed automated internal feedback system to reduce response latency

PROJECTS

Terminal UI Tetris Battle Royale

Jun 2026 - Present

[Project Description] *C*

- Building terminal based Tetris client that renders server-pushed game state in real time while handling non-blocking keyboard input and network I/O concurrently
- Implementing dependency-free Tetris engine as pure I/O-free logic, integrated against a custom encrypted protocol and authenticated TCP session such that server remains authoritative

JOM AI @ Tampines Hackathon 2026

May 2026

Python, TypeScript, React

- Achieved 2nd place overall, securing S\$2,400 in prize money as part of a 6-member team
- Developed a computer vision pipeline with Roboflow to detect illegally parked bicycles using CCTVs
- Built full-stack web-app featuring navigation, parking occupancy monitoring and dashboard visualisation

Beta CPU Tetris on FPGA

Jan 2026 - Apr 2026

Lucid, Verilog, Beta Assembly

- Rebuilt Tetris (1989) in Beta Assembly based on MIT's Beta CPU Architecture deployed on an Alchitry Au FPGA development board
- Integrated time-domain multiplexing, reducing LUT usage by sharing RAM resources between the CPU and VGA driver
- Implemented core game mechanics including collision detection, piece rotation and line clearing in assembly
- Designed hardware interfaces including custom 640x480 VGA controller and dual-channel speakers

SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript, C, Lucid, Verilog, Beta Assembly

Frameworks & Libraries: PyTorch, Node.js, React

Tools & Platforms: Linux/Unix, Git, Docker